STATS 401 · FINAL PROJECT POSTER

Who Can Reach the Metro in Ten Minutes?

Visualizing pedestrian access to metro, buses and bazaars in Tashkent, Uzbekistan

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THE HEADLINE ANSWER

13.6%

of the analysed population lives within a **modelled 10-minute walk of metro access**: about **435,690 of 3,212,200** residents.



12 districts · official SIAT 2026-Q2 population · 800 m along the pedestrian network (10 min at 4.8 km/h). A model estimate, not observed travel.

Introduction

Tashkent's metro grew from 29 stations in 2015 to 50, but a line on a map is not access on foot. We built a reproducible walking-network model and a linked interactive website so **residents, commuters and planners** can see who can walk to the metro, and who relies on buses.

Research questions

1. What share of residents live within a 10-minute walk of metro access?
2. Which districts have the highest and lowest coverage?
3. Which districts depend mostly on buses?
4. Are dense districts relatively underserved?
5. How does bazaar access overlap with metro and bus?

METHOD

Python · pandas · GeoPandas · OSMnx · SciPy · D3.js v7 · MapLibre GL JS

STEP 1
Acquire open data

- **OpenStreetMap** via OSMnx + Overpass: walk network (111,386 nodes), 50 stations, 144 entrances, 2,163 bus stops, 83 bazaars
- **SIAT** official 2026-Q2 district totals; **WorldPop** 100 m grid
- Each run logs endpoint, time, counts, checksums

STEP 2
Clean & calibrate

- **13 → 12 districts**: no official population for Yangi Toshkent, so it is excluded rather than invented
- **56 → 50 stations**: Latin/Cyrillic duplicates merged
- WorldPop only weights cells **within** districts, rescaled to official totals

STEP 3
Route on the walk network



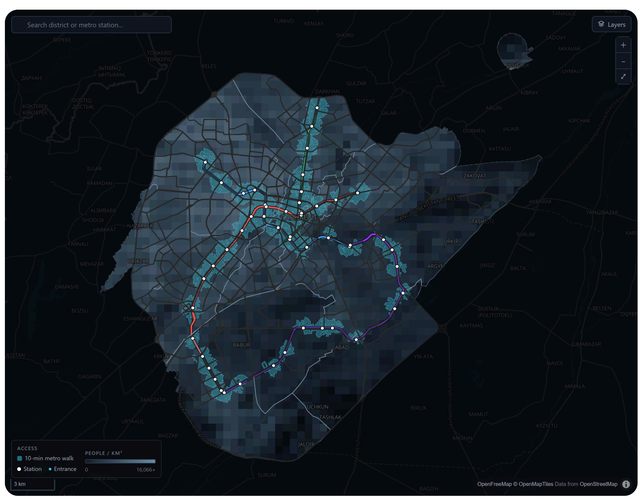
Network, not circles. 800 m is walked along streets: a barrier with one crossing (blue) cuts the reachable network (teal) well inside the straight-line circle (dashed). Multi-source Dijkstra from **154 metro access points** and every bus stop.

STEP 4
Classify & visualize

- **65,169 cells** → metro · bus-only · underserved, summed by district
- Sensitivity to walking speed and population surface
- Static site: D3 charts + MapLibre map sharing **one selection store**

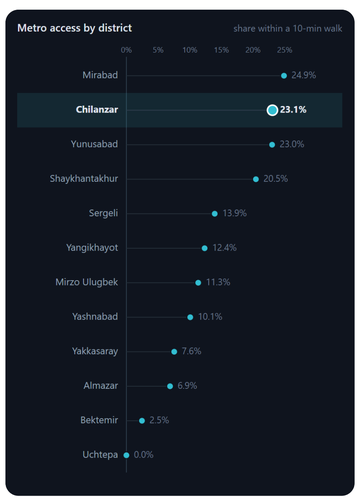
RESULTS · FIVE LINKED VIEWS

Screenshots of the live website; Chilanzar selected



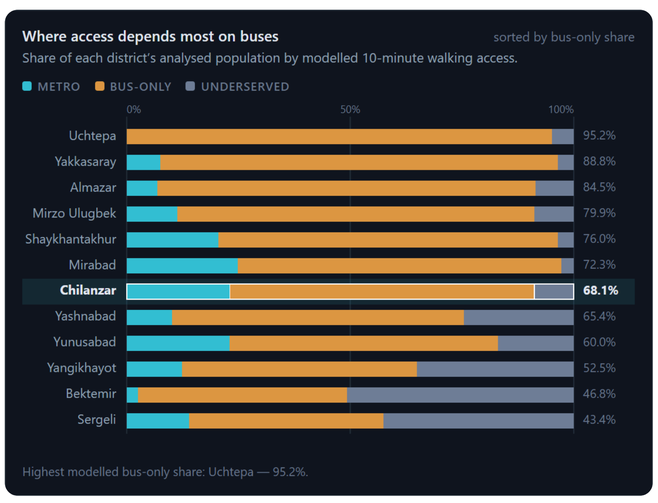
A Interactive access map Q1

Teal 10-minute metro catchment over population density, with bus-stop and bazaar layers, search and district selection. **Access is a narrow ribbon along the lines**; counting bus stops, 85.6% reach some transit.



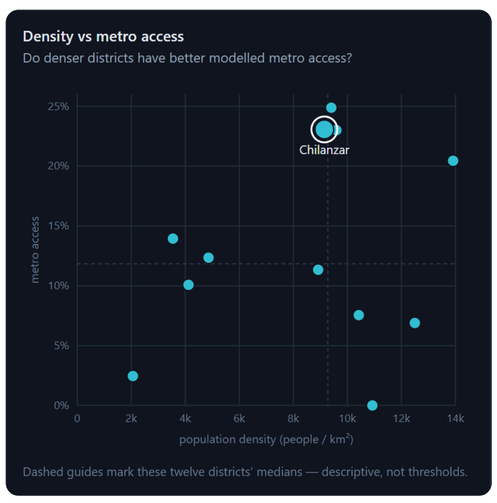
B Ranked dot plot Q2

Mirabad 24.9% → Uchtepa 0.0%. The top four hold 62% of all metro-accessible residents.



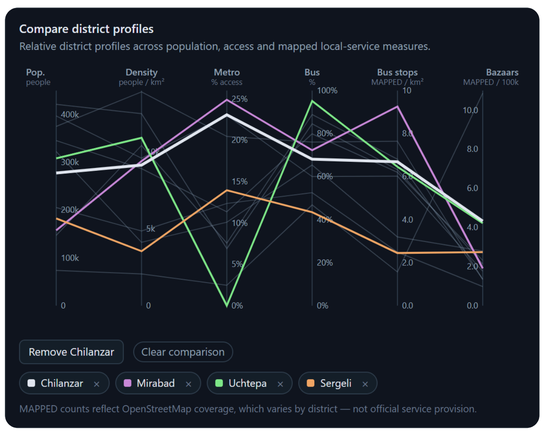
C 100% stacked composition Q3

Two different gaps. **Mostly bus-only:** Uchtepa 95.2%, Yakkasaray 88.8%, Almazar 84.5% bus-only. **Beyond both modes:** Bektemir 50.7%, Sergeli 42.6% underserved.



D Density vs access Q4

No relationship (Spearman ρ ≈ 0.01, n = 12). **Almazar, Uchtepa, Yakkasaray** are dense, below-median for metro, and mostly bus-only.



E Parallel coordinates COMPARE

Six attributes, up to four pinned districts. Profiles cross: Uchtepa is 0% metro but tops bus-only; Sergeli has the lowest bus-only share. **No district is uniformly strong or weak.**

Major interactions: one store links all five views

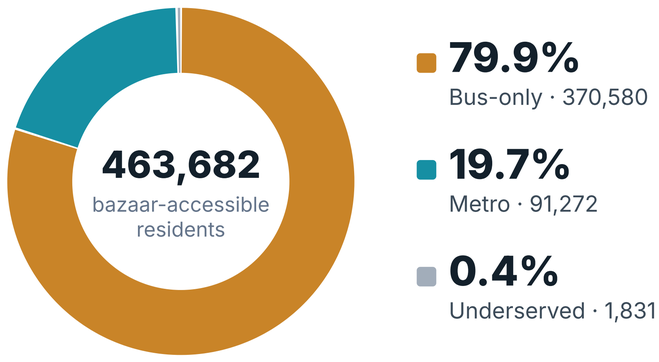
Hovering over Uchtepa in C highlights it in D and E and opens a tooltip.



- **Hover** highlights a district in every view
- **Click** selects it globally; the map zooms and the profile opens
- **Compare** pins up to four districts in colour
- **Search & layers** find a district or station; toggle buses and bazaars

Q5 · BAZAAR × TRANSIT OVERLAP

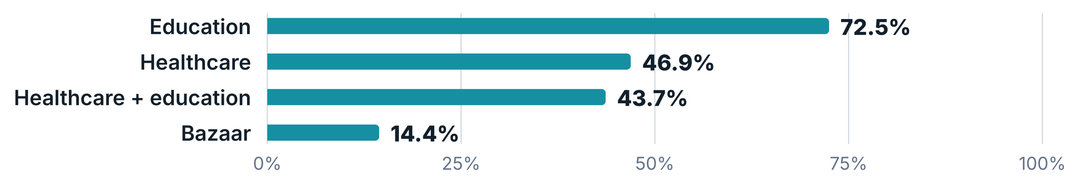
Four in five residents near a bazaar are bus-only



Residents within a modelled 10-minute walk of a mapped OSM marketplace, by transit class. Measures walking proximity, not shopping trips or transfers.

EVERYDAY DESTINATIONS · SAME CURRENT MODEL

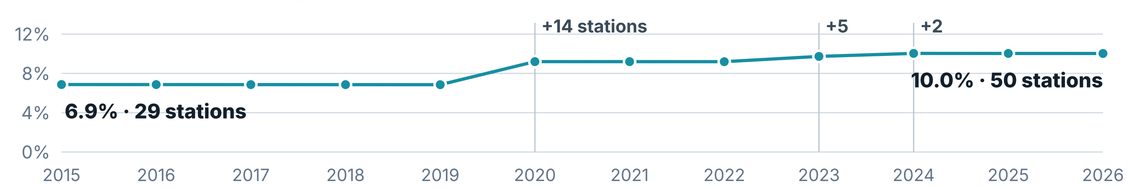
Schools are near most residents; bazaars are not



"Healthcare + education" is the true intersection of the two, not the sum of two shares.

STANDARDIZED 2015–2026 SERIES · SEPARATE METHOD

The 2020 station openings moved access the most



Station-centre proxies, fixed network, annual WorldPop weights: **not** comparable with the 13.6% headline.

Discussion

View	What (data)	Why (task)	How (marks · channels)
A Map	Catchment, density grid, stations, stops	Locate, explore	Areas, points, lines · luminance, teal hue
B Dot plot	12 districts × metro %	Rank, find extremes	Points · aligned position, sorted
C 100% bars	12 × three-class split	Part-to-whole	Segments · length, categorical hue
D Scatter	Density × metro %	Correlate, outliers	Points · x/y position, median guides
E Parallel	6 attributes per district	Compare profiles	Lines · position per axis, ≤ 4 highlights

Strengths

- Quantities use **position on common scales**
- **One colour code** across all views
- Linked **overview → details on demand**
- Uncertainty shown; keyboard and screen-reader access

Limitations

- **Modelled proximity**, not travel, frequency or fares
- District averages hide local variation (n = 12)
- OSM gaps; straight-line connectors; 10 stations lack entrances
- Parallel coordinates are unfamiliar to lay readers

How sensitive is 13.6% to walking speed?

Modelled metro share
at each tested walking speed



Conclusion

1. **About one in seven** residents (13.6%) can reach metro access in a modelled 10-minute walk.
2. Access concentrates in **four central districts** (20–25%); Uchtepa sits at 0.0%, only 19.7 m outside the threshold.
3. For **72%** of residents the walkable transit is a bus stop, most of all in Uchtepa, Yakkasaray and Almazar.
4. **Density does not predict access:** dense Uchtepa, Almazar and Yakkasaray rely mostly on buses; the low-density periphery lacks both.
5. Among residents who can walk to a bazaar, **79.9% are bus-only**.

> **What readers can take away:** residents can check whether the metro is realistically walkable from their district. Planners can tell apart two kinds of gap, dense districts that rely on buses and a periphery beyond both modes, as places to examine rather than verdicts on service quality.

Future work

Address search · walking-speed slider · mahalla-level aggregation · official bus-frequency data · lines under construction · testing with residents.

References

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Munzner, T. (2014). *Visualization Analysis and Design*. CRC Press.

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WorldPop (2025). Global 2015–2030 constrained population, R2025A. doi:10.5258/SOTON/WP00839

MapLibre GL JS v5.24 · OpenFreeMap / OpenMapTiles basemap.

Data & licences

OpenStreetMap contributors, ODbL: network, stations, entrances, stops, bazaars, facilities, boundaries.

SIAT, Statistics Agency of Uzbekistan, dataset 3890 (2026-Q2), CC BY 4.0, www.stat.uz

WorldPop R2025A 100 m, CC BY 4.0: within-district weights only.

Scan to explore the live website

tashkent-transit-walkability.saidkamolbahodirov.workers.dev

Code & data: github.com/Abdullajon1881/stats401-final-project